A Report

on

LOST & FOUND SYSTEM

carried out as part of the course Project Based Learning-3 (PBL-3)

Submitted by

Name of student

Roll no

V Semester

in partial fulfilment for the award of the degree

of

BACHELOR OF TECHNOLOGY

In

Computer Science & Engineering

Department of Computer Science & Engineering,
School of Computer Science and Engineering,
Manipal University Jaipur,

July-Dec 2025

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YASHVI TAUNK

23FE10CSE00634

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Under the Guidance of :

Guide Name : MRS.HARSHIKA MATHUR…….

Guide Signature(with date) : ……

Acknowledgement

This project would not have been completed without the help, support, comments, advice, cooperation and coordination of various people. However, it is impossible to thank everyone individually; I am hereby making a humble effort to thank some of them.

I acknowledge and express my deepest sense of gratitude to my internal supervisor ……Ms. Harshikha Mathur……. for his/her constant support, guidance, and continuous engagement. I highly appreciate his technical comments, suggestions, and criticism during the progress of this project titles "……LOST & FOUND SYSTEM…….".

I owe my profound gratitude to ***Dr. Neha Chaudhary*** , Head, Department of CSE, for her valuable guidance and for facilitating me during my work. I am also very grateful to all the faculty members and staff for their precious support and cooperation during the development of this project.

Finally, I extend my heartfelt appreciation to my classmates for their help and encouragement.

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MANIPAL UNIVERSITY
JAIPUR
(University under Section 2(f) of the UGC Act)

Department of Computer Science and Engineering

School of Computer Science and Engineering

Date: _____

CERTIFICATE

This is to certify that the project entitled "***(Project title)***" is a bonafide work carried out as PBL-3 ***End Term Assessment*** in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, by ***(YASHVI TAUNK)*** bearing registration number(_**23FE10CSE00634**), during the academic semester *VI of year 2024-2025.*

Place: Manipal University Jaipur, Jaipur

Name of the project guide: ___ MRS.HARSHIKA MATHUR _____

Signature of the project guide: _____

MINOR PROJECT PROGRESS REPORT GUIDELINES

COMPONENTS OF APPLICATION BASED PROJECT REPORT

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ABSTRACT

The Lost and Found System is a web-based platform developed to streamline the reporting, tracking, and retrieval of lost belongings within the college campus. Traditional methods such as manual registers, notice boards, and informal chat groups often result in unorganized data, delays in recovery, and difficulty in verifying ownership. This project introduces a centralized digital solution that allows users to report lost or found items, upload images, view potential matches, and claim belongings through an authenticated verification process.

The system is designed with an intuitive interface and a structured workflow comprising item reporting, automated matching, admin verification, and secure record storage. An SQL-based backend ensures reliable data handling, while the web portal improves accessibility for students and faculty. This report covers the system architecture, database design, DFDs, ER diagrams, implementation details, testing strategy, and expected outcomes. The Lost and Found System aims to reduce item loss incidents and ensure a smooth, transparent recovery process for the campus community.

1. INTRODUCTION

Many students frequently lose important items such as ID cards, wallets, earphones, books, and stationery within the college campus. The lack of a centralized, digital tracking system results in confusion, delays, and difficulty recovering belongings. Manual processes — including registers at security desks, WhatsApp groups, or posters — are inefficient, unstructured, and vulnerable to loss of information.

The Lost and Found System provides a unified digital platform where users can easily submit lost/found item reports, upload item images, search for matches, and claim items through verification. The system minimizes manual effort and ensures transparency in the recovery process while maintaining campus-wide accessibility. add one more paragraph in it

In addition to improving efficiency, the system enhances accountability by maintaining a structured digital record of every reported item. This reduces the chances of misinformation, duplicate entries, and disputes during the claim process. By enabling image uploads, category-based search, and admin verification, the Lost and Found System ensures that students can rely on a transparent and secure method for retrieving their belongings. The platform not only minimizes the workload of campus authorities but also promotes a more organized and user-friendly environment across the college.

1.1 Objectives of the Project

1. To create a centralized online platform for reporting lost and found items.
2. To store item details securely using a structured SQL database.
3. To enable users to upload images and search using keywords or categories.
4. To provide automated matching between lost and found entries.
5. To implement admin verification to prevent false claims.
6. To maintain complete records of recovered and pending items.

7. To improve campus efficiency and reduce delays in item retrieval.

1.2 Brief Description of the Project

The Lost and Found System is a web-based application designed to digitally manage lost and found items on the college campus. Users can log in, report items, upload descriptions and photographs, and check for matching records. When an item is found, the admin verifies the details and approves claims, ensuring that belongings are returned to legitimate owners. The system stores data in an SQL database, maintains logs, and provides an intuitive user interface for seamless access.

In addition to simplifying the process of reporting and tracking items, the system ensures a structured workflow that minimizes human error and eliminates dependency on traditional manual methods. Each item reported is categorized, stored, and time-stamped, enabling both users and administrators to quickly filter and locate records. By incorporating image uploads and keyword-based search, the system significantly enhances accuracy during item identification and matching.

Furthermore, the Lost and Found System introduces an admin-controlled verification layer to maintain the authenticity of claims. This prevents false reporting and ensures that every item is returned only after proper validation. With its user-friendly interface, secure database management, and smooth interaction flow, the system provides a dependable solution for campuses aiming to improve organization, transparency, and overall user satisfaction in lost-and-found operations.

1.3 Technology Used

1.3.1 Hardware Requirements

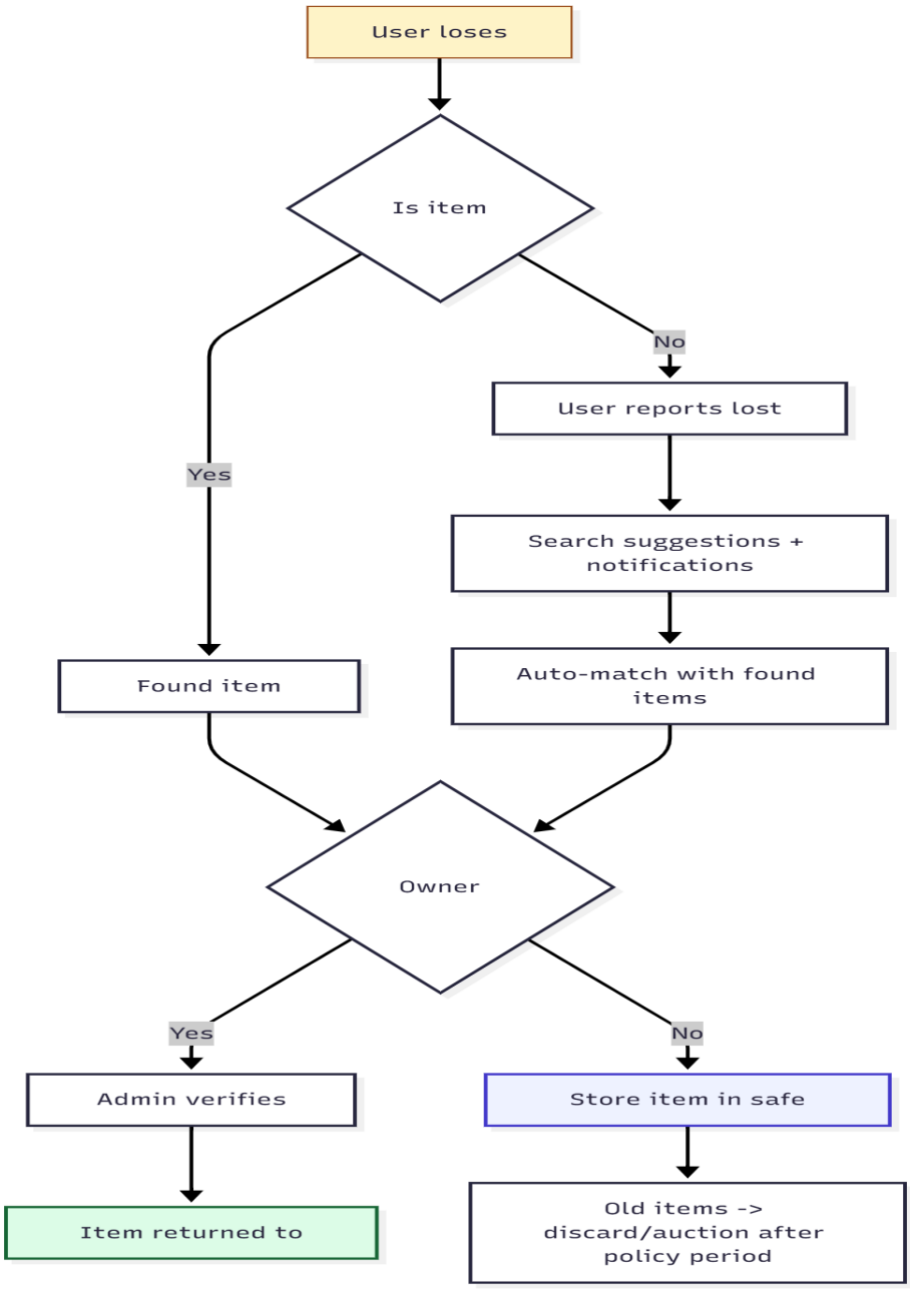
- Laptop/desktop with minimum **2 GB RAM** (recommended 4–8 GB)
- Stable internet connection
- 1 GHz processor or above
- Modern web browser (Chrome, Firefox, Edge)
- Minimum 128 GB storage
- Keyboard and mouse for smooth navigation
- Webcam (optional, for verification upgrades)
- Router/Wi-Fi device for stable connectivity

1.3.2 Software Requirements

- HTML, CSS, JavaScript for frontend
- PHP / Node.js / Java for backend
- SQL (MySQL / PostgreSQL) for database
- XAMPP / WAMP / MAMP for local server setup
- VS Code as primary code editor
- Git & GitHub for version control (optional)
- Bootstrap / Tailwind CSS for UI styling (optional)
- jQuery for enhanced frontend interactions
- phpMyAdmin / MySQL Workbench for DB management
- Postman for API testing
- JSON for structured data exchange
- Browser Developer Tools for debugging
- Figma / Canva (optional for UI mockups)

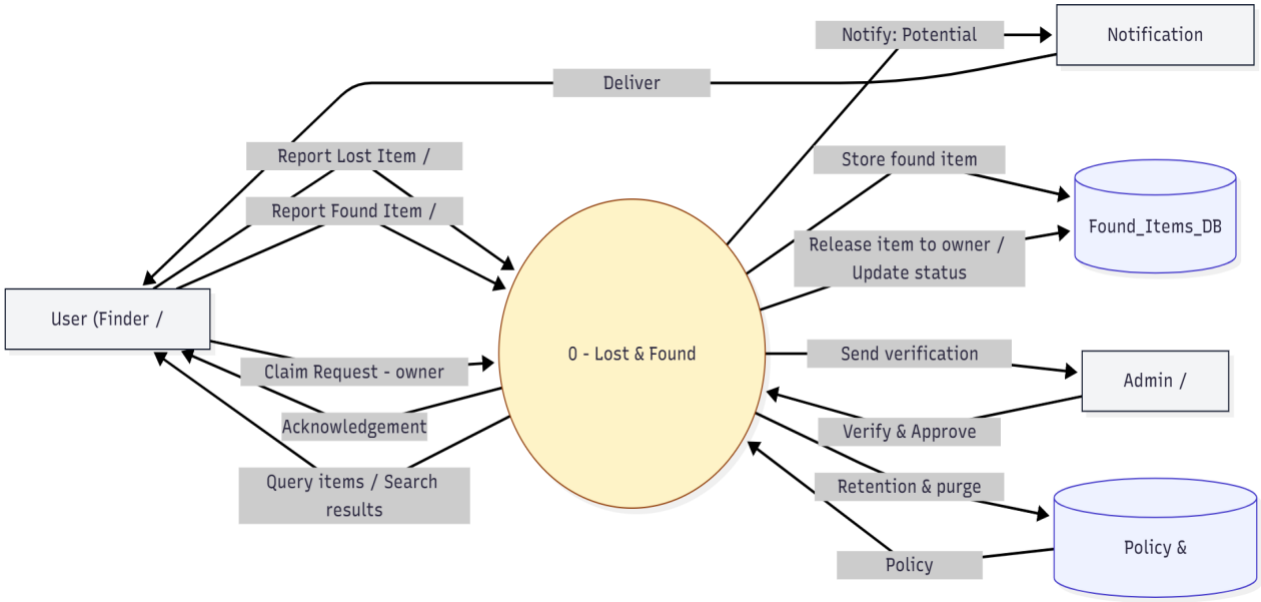
2. DESIGN DESCRIPTION-

2.1 Flow Chart

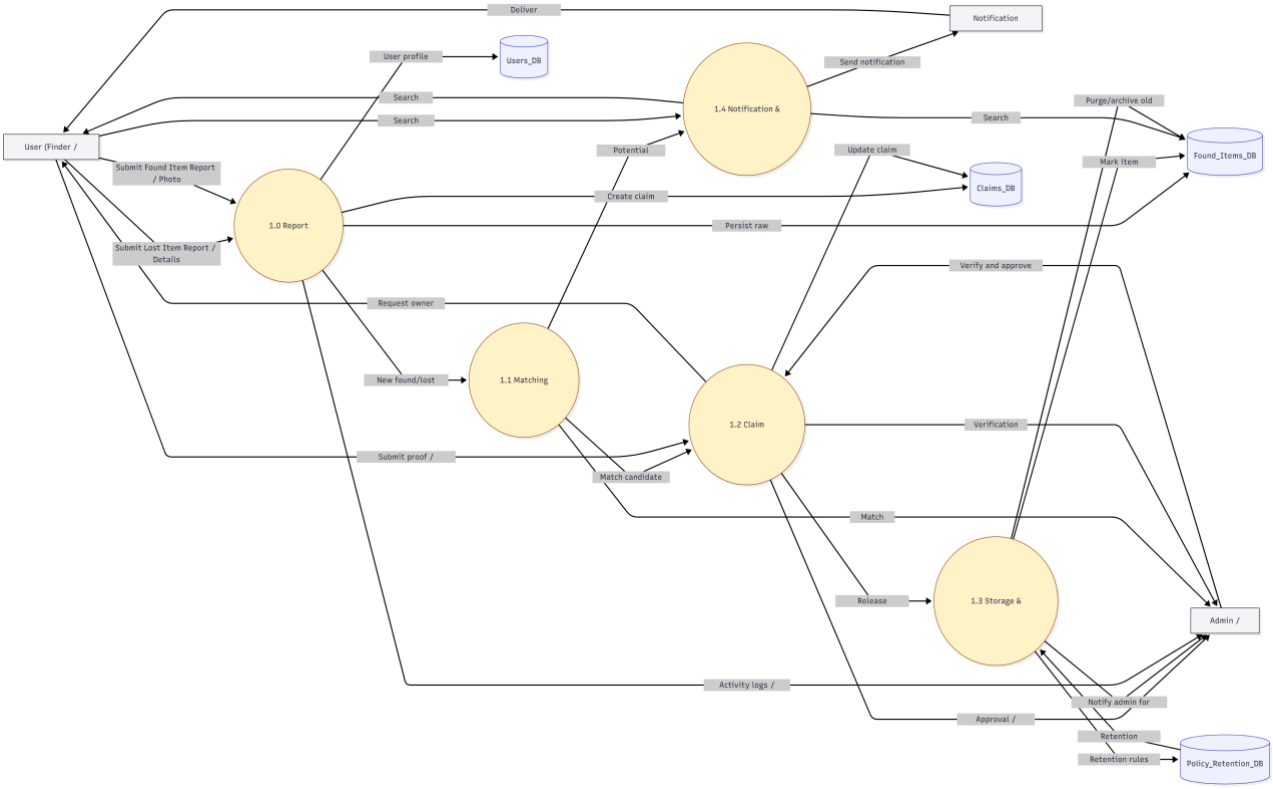


2.2 Data Flow Diagram (DFD)

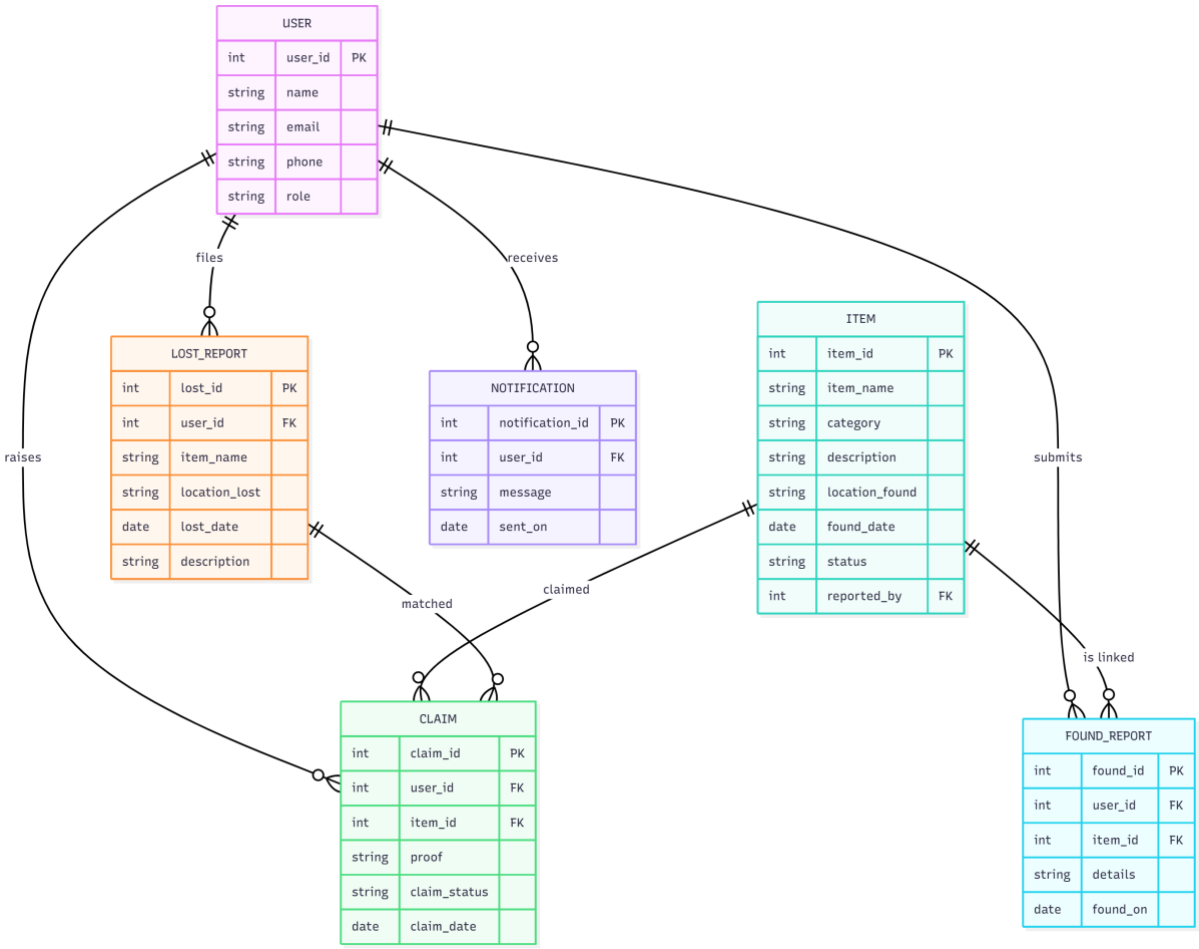
- **Level-0:**



Level 1:



2.3 Entity-Relationship Diagram (ERD)



3. PROJECT DESCRIPTION

3.1 Database

1. The system uses a relational database to store information about users, lost items, found items, claims, and notifications.

2. It ensures data accuracy and integrity using primary keys, foreign keys, and normalized tables.

3. The database supports fast searching of items based on name, category, and location.

4. Automatic matching between lost and found reports is done by comparing stored item attributes.

5. All data is securely stored and can be updated or verified by the admin when needed.

3.2 Table Description

- **Users Table**

Column	Description
user_id	Primary Key
name	Student Name
email	Email ID
phone	Contact number
role	user/admin

Items Table

Column	Description
item_id	Primary Key
type	Lost/Found
title	Name of item
description	Details
category	Category (ID card, wallet, books, etc.)
image	Image path
location	Where item was lost/found
date	Date reported
reported_by	User ID

Claims Table

Column	Description
claim_id	Primary Key
item_id	FK
user_id	FK
status	pending/approved/rejected
admin_id	FK
claim_date	Date of claim

3.3 File/Database Design

1. The database consists of multiple tables such as User, Item, Lost_Report, Found_Report, Claim, and Notification.

2. Each table is designed with fields, primary keys, and foreign keys to define relationships between entities.

3. Files (tables) store structured records like item details, user details, and claim information.

4. The design follows a normalized structure to reduce redundancy and ensure consistent data storage.

5. All tables support standard CRUD operations (Create, Read, Update, Delete) for smooth system functioning.

4. INPUT/OUTPUT FORM DESIGN

Input Forms

- Lost item submission form
- Found item submission form
- Admin verification panel
- User login/sign-up

Output Forms

- Matching item results
- Claim status updates
- Item database table
- Admin dashboard

5. TESTING & TOOLS USED

Testing

Unit Testing:

• Form validation (required fields, email format, image extension check)
• JavaScript-based matching logic
• Input sanitization to avoid invalid entries
• Button click events and UI behavior

Integration Testing:

• Connection between HTML frontend and JavaScript functions
• Data flow between frontend forms and backend PHP scripts (if used)
• Database connectivity through XAMPP (MySQL)
• Insert, update, delete operations for items and claims

System Testing:

- End-to-end workflow of reporting lost/found items
- Searching, filtering, and item matching
- Claim request submission and admin verification
- Image upload & retrieval testing
- Responsiveness across different devices (mobile/desktop)

User Acceptance Testing (UAT):

- Students testing reporting and searching features
- Admin testing verification, approval, and record management
- UI feedback to ensure system is easy to use
- Test cases based on real campus scenarios

Tools Used

- **VS Code** – main editor for HTML, CSS, JS
- **HTML5** – for creating the structure of all system pages
- **CSS3** – for styling, layout, and responsive design
- **JavaScript** – for client-side logic, validations, search, and user interaction
- **XAMPP** – running Apache server & MySQL for the backend database
- **phpMyAdmin** – managing database tables, queries, and relationships
- **Browser Developer Tools** – debugging layout, JS errors, and console logs
- **GitHub** (optional) – version control and backup
- **JSON** – for sending and receiving structured data between frontend and backend
- **Chrome/Firefox Browser** – testing and UI inspection
- **Postman** (optional) – API testing for any PHP endpoints if used

6. IMPLEMENTATION & MAINTENANCE

- Frontend built using HTML/CSS/JS
- Backend built using PHP/Node.js (your choice)
- SQL database for storing records
- Code easily maintainable and extensible
- Periodic data cleanup
- Security updates for login system

7. CONCLUSION & FUTURE WORK

The Lost and Found System provides a structured and efficient platform to manage lost and found items within the campus. It eliminates the inefficiencies of manual processes and allows students to recover items faster through a transparent and reliable system.

Future Enhancements

- AI image-based matching
- QR code tagging of items
- Mobile app
- Notification alerts via email/SMS
- Role-based access for departments

8. OUTCOME

- Fully functional Lost and Found portal
- Structured SQL database
- User login and reporting module
- Admin verification and tracking
- DFD, ERD, and flow diagrams

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